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Systems, Roles, and Development Methodologies

Systems Analysis and Design, 8e
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System

- A set of components that work together to achieve a common goal.
- A group of interrelated components working together towards a common goal by accepting input and producing output in an organized transformation process.
- Components of a System

There are three basic components of a system, they are

 - Input: involves capturing and assembling elements that enter to the system to be processed. Some of the inputs are raw materials, data etc.
 - Processing: It involves transformation process that converts input to output.
 - Output: It involves transforming element that has been produced by a transformation process to their ultimate destination.



Types of System

- Dynamic System: When the interrelated component of the system interacts with each other.
- Cybernative System: Dynamic System implementing the concept of feedback and control .
- Open System: A system got interacts with other system in its environment.
- Closed System: Stand-alone system that has no contact with other systems.
- Adoptive System: A System having the ability to change itself and its environment in order to survive.



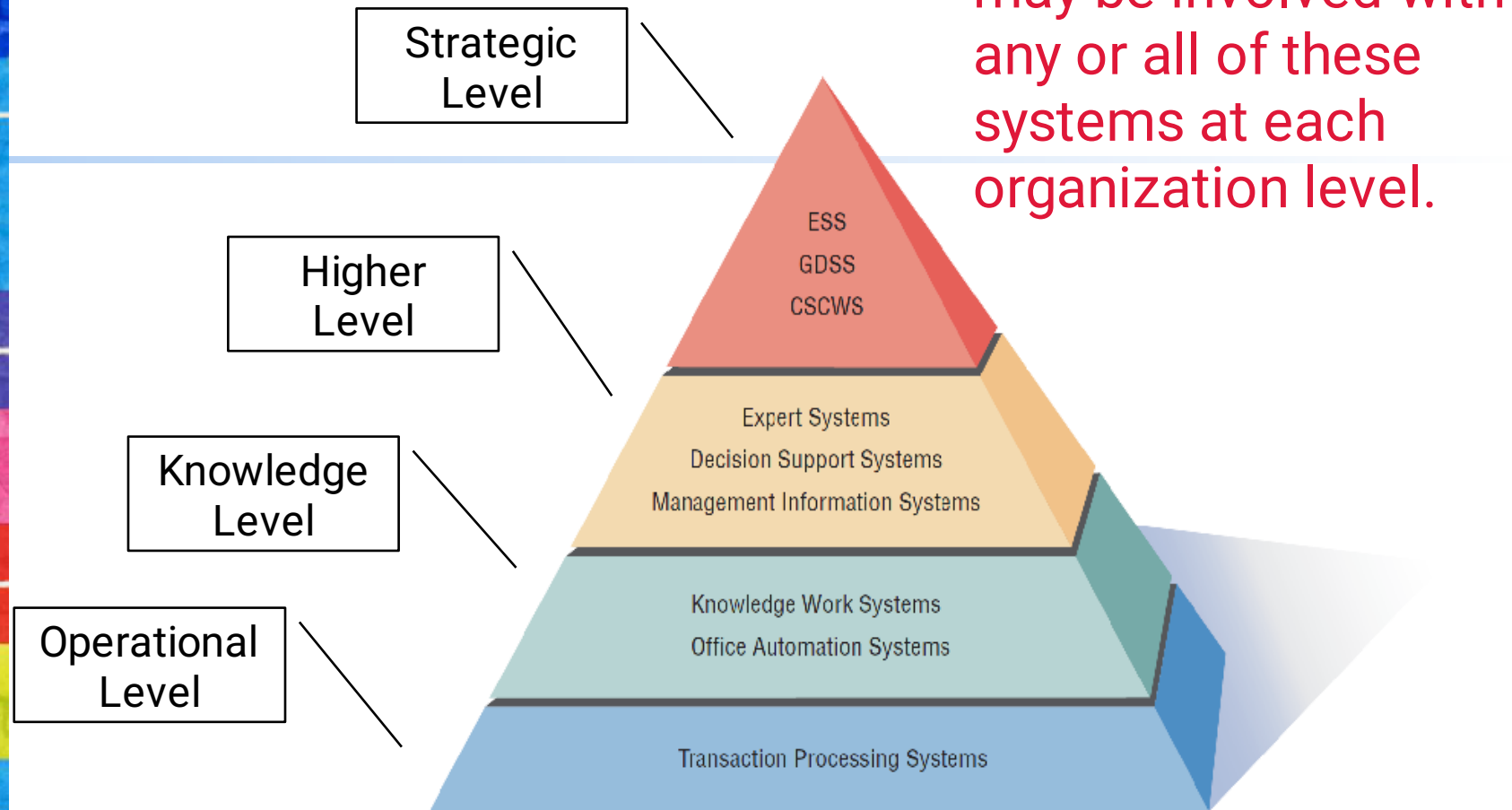
Information System

- An information system can be defined as a set of interrelated components that collect (or retrieve), process, store and distribute information to support decision making, coordination and control in an organization.
- Components of an Information System
 - Data: input that the system takes to introduce information.
 - Hardware: A computer and its peripheral equipment: input, output, and storage devices.
 - Software: refers to computer programs and the manuals (if any) that support them.
 - Telecommunication: Hardware and software that facilitate transmission and reception.
 - People: Information systems professionals and users.
 - Procedures: Rules for achieving optimal and secure operations in data processing.



Systems Analysts Recommend, Design, and Maintain Many Types of Systems for Users

- Transaction Processing Systems (TPS)
- Office Automation Systems (OAS)
- Knowledge Work Systems (KWS)
- Management Information Systems (MIS)
- Decision Support Systems (DSS)
- Expert Systems (ES)
- Executive Support Systems (ESS)
- Group Decision Support Systems (GDSS)
- Computer-Supported Collaborative Work Systems (CSCWS)





Operational Level

- Transaction Processing System (TPS)
 - Process large amounts of data for routine business transactions
 - Boundary-spanning
 - Support the day-to-day operations of the company
 - Examples: Payroll Processing, Inventory Management



Knowledge Level

- Office Automation System (OAS)
 - Supports data workers who share information, but do not usually create new knowledge
 - Examples: word processing, spreadsheets, desktop publishing, electronic scheduling, communication through voice mail, email, teleconferencing
- Knowledge Work System (KWS)
 - Supports professional workers such as scientists, engineers, and doctors
 - Examples: computer-aided design systems, virtual reality systems, investment workstations



Higher Level

- Management Information System (MIS)
 - Supports a broad spectrum of organizational tasks including decision analysis and decision making
 - Examples: profit margin by sales region, expenses vs. budgets
- Decision Support System (DSS)
 - Aids decision makers in the making of decisions
 - Examples: financial planning with what-if analysis, budgeting with modeling
- Expert System (ES) and Artificial Intelligence
 - Captures and uses the knowledge of an expert for solving a particular problem which leads to a conclusion or recommendation
 - Researching understanding natural language and the ability to reason through a problem to its logical conclusion



Strategic Level

- Executive Support System (ESS)
 - Helps executives to make unstructured strategic decisions in an informed way
 - Examples: drill-down analysis, status access
- Group Decision Support System (GDSS)
 - Permit group members to interact with electronic support.
 - Examples: email, Lotus Notes
- Computer-Supported Collaborative Work System (CSCWS)
 - CSCWS is a more general term of GDSS.
 - May include software support called *groupware* for team collaboration via network computers
 - Example: video conferencing, Web survey system



Need for Systems Analysis and Design

- Installing a system without proper planning leads to great user dissatisfaction and frequently causes the system to fall into disuse.
- Lends structure to the analysis and design of information systems
- A series of processes systematically undertaken to improve a business through the use of computerized information systems



Roles of the Systems Analyst

- The analyst must be able to work with people of all descriptions and be experienced in working with computers.
- Three primary roles:
 - Consultant
 - Supporting expert
 - Agent of change



Qualities of the Systems Analyst

- Problem solver
- Communicator
- Strong personal and professional ethics
- Self-disciplined and self-motivated



Systems Development Life Cycle (SDLC)

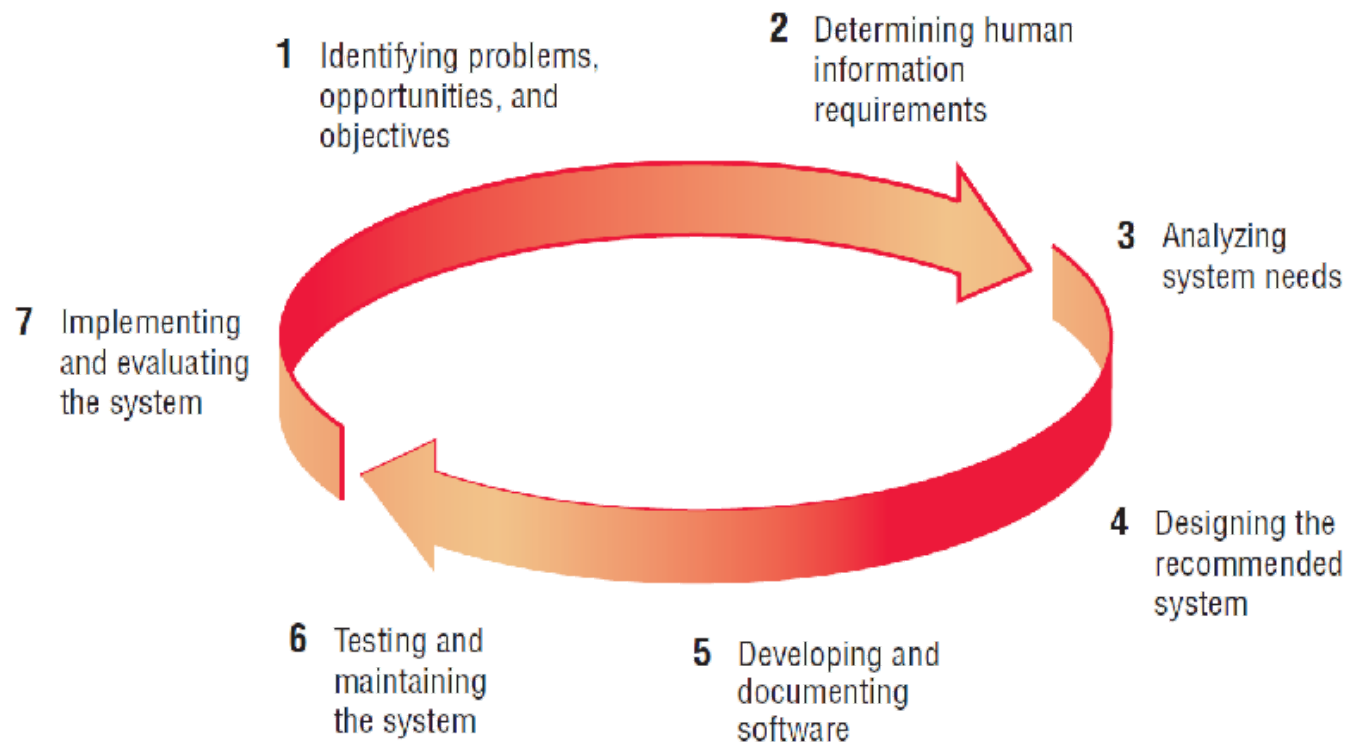
- SDLC: is the process of understanding how an information system (IS) can support business needs by designing a system, building it, and delivering it to users.



Systems Development Life Cycle (SDLC)

- The systems development life cycle is a phased approach to solving business problems.
- Developed through the use of a specific cycle of analyst and user activities
- Each phase has unique user activities.

The Seven Phases of the Systems Development Life Cycle (Figure 1.3)





Identifying Problems, Opportunities, and Objectives

- Activity:
 - Interviewing user management
 - Summarizing the knowledge obtained
 - Estimating the scope of the project
 - Documenting the results
- Output:
 - Feasibility report containing problem definition and objective summaries from which management can make a decision on whether to proceed with the proposed project



Determining Human Information Requirements

- Activity:
 - Interviewing
 - Sampling and investigating hard data
 - Questionnaires
 - Observe the decision maker's behavior and environment.
 - Prototyping
 - Learn the who, what, where, when, how, and why of the current system.
- Output:
 - The analyst understands how users accomplish their work when interacting with a computer; and begin to know how to make the new system more useful and usable. The analyst should also know the business functions and have complete information on the people, goals, data, and procedure involved.



Analyzing System Needs

- Activity:
 - Create data flow, activity, or sequence diagrams.
 - Complete the data dictionary.
 - Analyze the structured decisions made.
 - Prepare and present the system proposal.
- Output:
 - Recommendation on what, if anything, should be done



Designing the Recommended System

- Activity:
 - Design procedures for data entry.
 - Design the human-computer interface.
 - Design system controls.
 - Design database and/or files.
 - Design backup procedures.
- Output
 - Model of the actual system



Developing and Documenting Software

- **Activity:**
 - System analyst works with programmers to develop any original software.
 - Works with users to develop effective documentation.
 - Programmers design, code, and remove syntactical errors from computer programs.
 - Document software with help files, procedure manuals, and Web sites with Frequently Asked Questions.
- **Output:**
 - Computer programs
 - System documentation



Testing and Maintaining the System

- Activity:
 - Test the information system.
 - System maintenance.
 - Maintenance documentation.
- Output:
 - Problems, if any
 - Updated programs
 - Documentation



Implementing and Evaluating the System

- Activity:
 - Train users.
 - Analyst plans smooth conversion from old system to new system.
 - Review and evaluate system.
- Output:
 - Trained personnel
 - Installed system